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Supplementary Examination

Course: B. Tech. Branch: Chemical Engineering Semester: VII

Subject: Process Dynamics and Control (BTCHC701)

Max Marks: 60 Date: 13/08/2022

Instructions to the Students:

1. *All the questions are compulsory.*
2. *Assume suitable data wherever necessary and mention it clearly.*

Q. 1 **A** A thermometer having a time constant of 0.2 min. is placed in a temperature bath and after the thermometer comes to equilibrium with the bath, the temperature of the bath is increased linearly with time at a rate of 1^{o} /min. What is the difference between bath temperature and indicated temperature at i) 0.1 min. ii) 1 min. after the change in temperature begins. **(6 M)**

B Differentiate between feedback and feed forward control strategies. Develop a feedback control and feed forward control systems for controlling the temperature of liquid inside the stirred tank, which is heated by steam passing through the steam coil immersed inside the tank. Identify the different variables involved in the process. **(6 M)**

Q. 2 Solve any *TWO*.

A Develop State Equations for stirred tank heater and mention the different variables used for the same. **(6 M)**

B Estimate the time response of Ramp Input for first order system. **(6 M)**

C What are the various forcing functions used to study the dynamic response of a system? Explain each with neat sketch. **(6 M)**

Q. 3 The overall transfer function of the control system is given as, **(12 M)**

$G(s) = \frac{1}{1.5s^2+2.4s+6}$ A step change of magnitude 6 is introduced into the system. Determine the following

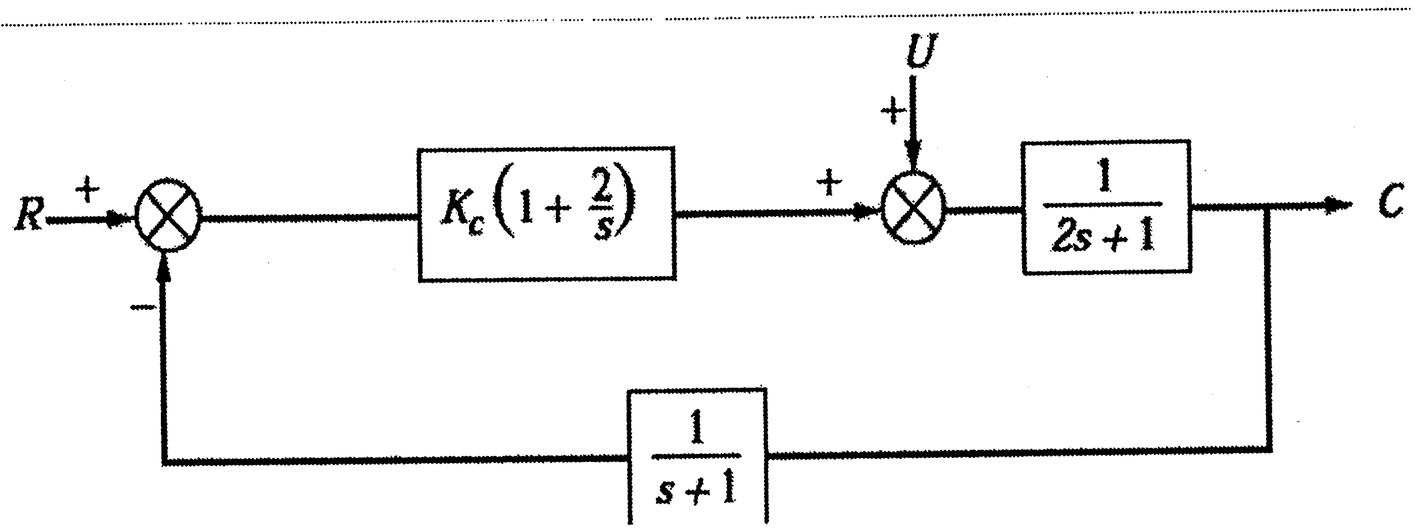
a. Overshoot
b. Period of Oscillation
c. Decay ratio
d. Rise time
e. Ultimate value of response
f. Maximum value of response

Q. 4 **Solve any *TWO*.**

A Derive the expression for Amplitude Ratio and phase angle for pure capacitive system and also plot asymptotic Bode diagram for the same. **(6 M)**

B Explain in detail how location of poles on the complex plane affects the stability of the system **(6 M)**

C Write the characteristic equation for the control system shown in Figure .Use the Routh test to determine if the system is stable for Kc=4, Determine the ultimate value of Kc above which the system is unstable. **(6 M)**



Q. 5 Explain the root locus and its significance. Plot the root locus of the system having transfer function $G_p(s) = \frac{1}{(s+1)(s+2)}$ and consider $G_m(s) = G_f(s) = 1$. **(12 M)**

OR

Q. 5 **A** Starting from the material balance, Develop the overall transfer function for non-interacting system. (Assume the necessary data and mention it). **(6 M)**

B Derive the expression for Amplitude Ratio and phase angle for pure capacitive system and also plot asymptotic Bode diagram for the same. **(6 M)**

**********End**********